Cybersecurity Basics for Students

Knowledge document 008 of 100

Introduction to Cybersecurity

Cybersecurity is the practice of protecting computers, networks, mobile phones, software and data from unauthorised access, theft, damage or disruption. Students use the internet daily for study, banking apps, social media and games, which makes them targets for criminals. Cybersecurity rests on three goals known as the CIA triad: Confidentiality (only authorised people can see information), Integrity (information is accurate and not secretly changed) and Availability (systems and data are accessible when needed). Good habits, not only technology, are the best defence, because most attacks succeed by tricking people.

Common Cyber Threats

Malware is malicious software, including viruses, worms, trojans, spyware and ransomware. Ransomware locks or encrypts a victim's files and demands payment to restore them; the WannaCry attack of May 2017 affected hundreds of thousands of computers in about 150 countries. Social engineering manipulates people into giving away information. A data breach exposes personal data. A DDoS attack floods a website with traffic to make it unavailable. Identity theft and online financial fraud, such as fake loan apps, fake job offers, lottery scams and "digital arrest" scams where fraudsters impersonate police or officials on video calls, are widespread in India.

What Is Phishing?

Phishing is a type of cyber attack in which criminals pretend to be a trusted person or organisation, such as a bank, a delivery company, a government office or a college, to trick victims into revealing sensitive information like passwords, OTPs, card numbers or into clicking harmful links. Phishing usually arrives by email, but variants include smishing (through SMS), vishing (through voice calls) and spear phishing (targeting a specific person with personalised messages). Warning signs include urgent or threatening language ("your account will be blocked today"), spelling mistakes, an unfamiliar sender address, links that do not match the real website and requests for OTP or password. A bank will never ask you for your OTP or PIN. To stay safe, do not click suspicious links, check the web address, never share OTPs and report the message.

What Makes a Password Strong?

A strong password is long, unpredictable and unique. Experts recommend at least 12 to 16 characters, mixing uppercase and lowercase letters, numbers and symbols, and avoiding personal details like names, birth dates or phone numbers, as well as common choices like "123456" or "password". A passphrase of several random words, such as a short unusual sentence, is both strong and easy to remember. Every account should have a different password, so that one leak does not expose everything. A password manager can create and store strong unique passwords safely. Passwords should never be shared and should be changed if a breach is suspected.

What Is Two-Factor Authentication?

Two-factor authentication (2FA), also called multi-factor authentication (MFA), is a security method that requires two different kinds of proof of identity to log in. The factors are something you know (a password or PIN), something you have (a phone, a security key or an authenticator app) and something you are (a fingerprint or face). For example, after typing your password you must also enter a one-time code sent to your phone or generated by an authenticator app. Even if a criminal steals your password, they cannot log in without the second factor. Authenticator apps and hardware keys are safer than SMS codes, which can be intercepted by SIM-swap attacks. 2FA should be enabled on email, banking and social media accounts.

Other Safe Practices

Keep the operating system and apps updated, because updates fix security holes. Install apps only from official stores. Use antivirus software and a firewall. Avoid public Wi-Fi for banking, or use a VPN. Back up important files regularly, following the 3-2-1 rule: three copies, on two types of media, with one stored off-site. Lock your phone, and think before sharing personal information on social media. HTTPS and the padlock symbol in the browser show an encrypted connection, but they do not prove a website is honest.

Cybersecurity in India

India's national agency for cyber incidents is CERT-In, the Indian Computer Emergency Response Team, set up in 2004 under the Ministry of Electronics and Information Technology. The Information Technology Act, 2000 is India's main cyber law, and the Digital Personal Data Protection Act, 2023 governs personal data. Victims of online financial fraud should call the national cybercrime helpline 1930 immediately and report on the portal cybercrime.gov.in; quick reporting can help freeze the fraudsters' money. The Indian Cyber Crime Coordination Centre (I4C) coordinates law enforcement. October is observed as Cyber Security Awareness Month.

Challenges and Future

Attackers now use AI to write convincing phishing messages and deepfake voices and videos, while defenders use AI to detect threats. Growing numbers of connected devices expand the attack surface. Careers in ethical hacking, security analysis and digital forensics are growing quickly.

Quick Facts

- Phishing tricks people into revealing passwords, OTPs or money by pretending to be a trusted source.
- A strong password is long (12 or more characters), unique and unpredictable.
- Two-factor authentication needs two different proofs of identity, such as a password plus a one-time code.
- The CIA triad stands for Confidentiality, Integrity and Availability.
- In India, report online financial fraud on helpline 1930 and at cybercrime.gov.in.